

DATA PIPELINE (Sections 3.2 and 3.5)

Collection

6 typhoon events
2020–2024

Cleaning

Remove duplicates
Normalize text

Privacy Scrubbing

Remove identifiers
Anonymous post IDs

Language Groups

Tagalog / English
Three CMI strata

Annotation

Two annotators
Three triage tasks

LODO Folds

6 held-out events
15% dev split

Corpus Store

Frozen labels and splits
Identical fold inputs

MODEL CORE SUBSYSTEM (Section 3.8.1)

SEA-LION-ModernBERT-300M Shared Encoder

Gemma-3 SentencePiece Vocabulary
($V = 262,144$), Dimension $d = 768$
22 ModernBERT Layers with Rotary Position Embeddings (RoPE)
Input Embedding Footprint: 2.01×10^8 parameters (805 MB in FP32)
Shared pretrained encoder for all three tasks

NER Token Classification Head

Linear token projection to
BIO location/infra spans

Intent Classification Head

Sequence mean-pooling to 8
humanitarian intent classes

Urgency Assessment Head

Sequence mean-pooling
to 3 priority levels

TRAINING ORCHESTRATOR (240 Runs)

- **8 Configurations:** Three single-task baselines and five multi-task methods
- **LODO Partitioning:** $K = 6$ held-out disaster folds
- **Seeds:** $S = 5$ per fold (30 runs per configuration)
- **Run Integrity:** Loss divergence checks and automated restart

SERVING ORCHESTRATOR (Profiling)

- **Shared Multi-Task Model:** One encoder and three task heads
- **Sequential Condition:** Three single-task baselines run separately
- **Concurrent Condition:** Three heads run on the shared encoder
- **Trials:** 100 timed batch-1 trials

RESULTS AND TELEMETRY OUTPUTS (Section 3.8.1)

- **Execution Manifest (JSON):** Pinned environment, git hash, library versions, seeds, and hyperparameter grids
- **Training Trajectory Logs (CSV):** Step-level task losses, validation Macro-F1 scores, and early stopping events
- **Checkpoint Evaluation Registry (CSV):** Per-checkpoint out-of-domain Macro-F1 across holdout disaster events
- **Hardware Profiling Logs (CSV):** Millisecond latency, peak GPU VRAM, and host resident RAM per trial